Dragonfly URS Tender issue v1.1

Document Information

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1.01 The Requirement

This tender concerns the installation of automation at a warehouse operated by Firefly for its customer Firefly[1]. Currently, all pallet movements are handled by manually operated MHE (VNA, FLT, PPT) at the site[2]. The aim is to reduce operational costs and provide a resilient solution to the design data, with identified headroom and solutions for future volume increases[3]. This ITT invites proposals based on the design throughput information and building information[4].

1.02 The Site

The warehouse is 43,860 m² (472,000 sq ft) in size with VNA adjustable pallet racking (APR) installed, providing approximately 54,000 pallet positions[5]. The site operates 22 hours per day, 364 days per year[6].

A drawing titled "XYZ drawing for tender 22_07_24" shows the APR details and the internal site layout[7]. The site is available for dimension surveys by appointment[8].

1.03 Tender & Finance

The tender process will be managed by Firefly, with key stakeholder support from its customer[9]. Automation providers are invited to tender solutions to the outlined operational design[10]. Each tenderer must confirm they can support the requirements[11].

The financing and capital investment process needs to be confirmed as to whether Firefly or its customer will be financing the automation implementation[12]. All tenders with full costs must be supplied by September 27, 2024[13]. A turnkey solution is required[14]. Proposals should satisfy both the average and peak day site volumes[15].

Additional Requirements:

- Lease cost options should also be provided[16].
- Lead times to go live should be provided[17].
- Maintenance and repair proposals should be detailed, with full package options offered[18].
- Implementation, project, start-up, and spares costs should be itemized[19].
- The onsite WMS is Reflex, and the supplier is expected to interface with it for all associated warehouse, stockkeeping, and customer order fulfillment[20].

- CA certification is required, including compliance with the latest standards for driverless industrial trucks and their systems, such as ISO 3961-4:2023[21].
- Fire Alarm systems must be integrated to the existing system and stop in the event of an alarm[22].
- Full life cycle costs for a period of 10 years should be submitted[23].
- Ongoing support and software costs must be detailed[24].

Tender Contacts:

All queries and communications should be addressed to the designated Firefly contacts by email.

1.04 Automation expectation and requirements

The following functional areas are the subject of this ITT:

Phase 1

- **i.** Automate the unloading of pallets from the trailer (with consideration for options like auto loading/unloading with conveyors)[25].
- **ii.** Automate the movement of pallets from arrival at the ground floor Marshall location to the VNA P&D locations[26].
- **iii.** Automate the movement of pallets from the P&D location to floor locations marshalled at the despatch dock door, prior to truck loading, and also to the aisle 60 locations[27].

Phase 2

- **i.** Automate the movement of pallets from the P&D location to the storage putaway in the VNA APR storage location[28].
- **ii.** Automate the movement of pallets from the VNA APR storage location to the P&D location[29].

Firefly will consider proposals in these phases, with the implementation order to be confirmed[30]. Cost proposals for automating pallet unloading/loading (Phase 1, section i) should be itemized and separate from other proposals[31].

1.05 Documentation

The following documents form the basis of the tender agreement:

- Move_Detail-25th May 2022-Peak Volume.xlsx (Throughput volume file)[33].
- Move_Detail-30th May 2024-Average Daily Volume.xlsx (Throughput volume file)[34].
- Finished Product Pallet Labelling Requirements.docx (Firefly document)[35].
- Firefly Materials handling equipment mechanical plant and services v1.1.pdf (Mechanical standards document)[36].
- Firefly MHE testing standards v1.1.pdf (MHE testing standards document)[37].
- Master Purchase Agreement Firefly Log Europe.docx (Purchase Contract agreement)[38].

1.06 Design (throughput) data to be considered

The data to be used for the automation design is shown in the attached files for peak day volume (25th May 2022) and average day volume (30th May 2024)[39].

- Peak daily volume is 161 loads, or 7.3 loads per hour[40].
- The proposal should include a solution that can reach 9 loads per hour during a peak emergency period of two hours[41].

Peak day Volume 25th May 2022

Row Labels - Count of MVNSUP	
EFLT	**2964**
Inbound (with interface 1 scan)	460

Other (without interface upto 3 scans)	637
Outbound	1734
Replenishment (also halfs and to aisle 60 area)	133
PPT	**1340**
Inbound (with interface 1 scan)	338
Other (without interface upto 3 scans)	493
Outbound	479
Replenishment (also halfs and to aisle 60 area)	30
EVNA	**4202**
Inbound (with interface 1 scan)	874
Other (without interface upto 3 scans)	1083
Outbound	2074

Replenishment (also halfs and to aisle 60 area)	171
Grand Total	**8506**

Marshalling Tasks

Type	Moves
Inbound	460
Inbound	637
Out	1734
Internal	133
Inbound	338
Inbound	493
Out	479
Internal	30
Total Out	2213

Total In	1928
Total Internal	163
Total Moves	4304
Total Out Trailers	85.12
Total In Trailers	74.15
Total Trailers	159.27

Average day Volume 30th May 2024

Row Labels - Count of MVNSUP	
FLT	**2145**
Inbound	304
Other	203
Outbound	1541
Replenishment	97

PPT	**1167**
Inbound	336
Other	470
Outbound	335
Replenishment	26
VNA	**3199**
Inbound	624
Other	677
Outbound	1768
Replenishment	130
Grand Total	**6511**

Marshalling Tasks

Type	Moves

Inbound	304
Inbound	203
Out	1541
Internal	97
	1167
Inbound	336
Inbound	470
Out	335
Internal	26
	3199
Total Out	1876
Total In	1313

Total Internal	123
Total Moves	3312
Total Out Trailers	72.2
Total In Trailers	50.5
Total Trailers	122.7

1.07 Pallet load description - dims and tolerances

The pallets in use are wooden UK LPR/Chep type and Euro type[42].

UK IND PALLET FOR FINISHED GOODS AND MATERIALS

Pallet type	Firefly Requirement
Pl = length mm	1200
Pw = width mm	1000
Ph = pallet height mm - With Stock	2850
Pdw = pallet deflection, width mm	10

Pdl = pallet deflection, length mm	10

Pallet material	Wood & Plastic
Max. pallet weight kg	29
L = TU length max. mm	1210
W1= TU width max. mm	1010
Dl = Overhang (length) mm	10
Dw = Overhang (width) mm	10
Max Loaded weight (KG) (Footprint)	1000
Typical Weight Average (KG) (Footprint)	350

Note - Overhang relates to product lean only and not intentional pallet stacking overhang[43]. Pallet weights stated are the maximum expected for their pallet type, but many products will have a lower load weight[44].

EURO PALLET FOR FINISHED GOODS

Pallet type	Firefly Requirement

Pl = length mm	1200
Pw = width mm	800
Ph = pallet height mm - With Stock	2850
Pdw = pallet deflection, width mm	10

1.08 Description of the site current operation

Operational Process July 2024

- 1. Dock door layout:

  The main side of the warehouse operates between bays 1-35[45]. The other side of the warehouse has bays 36-70, but only four are operational for loading, unloading, and ad hoc activities[46].
- 2. Marshalling Bay layout:

  There are 24 marshalling bays opposite Bays 3 to 33[47]. Each bay has three rows and can accommodate nine pallets[48]. All bays are dynamic and can be used for either inbound or outbound activity[49].
- 4. Process Inbound:

  Inbound receipts are currently handled in three ways[50]. 60% of all inbound freight is "With Interface," which means SSCCs are on the system and require a single scan[51]. 40% of inbound freight is "Without Interface," which requires a three-scan process[52]. Unplanned or planned returns also require a three-scan process[53].
- 6. Put Away Logic:
  All storage locations can receive all products, from "Very Fast Moving" to "Slow Moving"[54].

- 7. P & D Process:

  All tasks are allocated by the Reflex Warehouse Management System (WMS)[55].

1.10 Description of the new automated proposed operation

Separate proposals are required for Phase 1 and Phase 2[56].

Phase 1 (Unloading/loading, transport to/from marshalling to P&D, and added value to aisle 60)

- **Stage (i):** Automate the unloading of pallets from trailers, particularly those with internal conveyors[57575757]. Auto scanning of the pallet label is required at this stage for pallet receipt[58].
- **Stage (ii):** Automate the movement of pallets from the ground floor Marshall location to the VNA P&D locations[59]. This stage should broadly support the movement of 95 pallets per hour to the P&D locations[60].
- **Stage (iii):** Automate the movement of pallets from the P&D location to floor locations at the despatch dock door, prior to truck loading[61]. This stage should broadly support the movement of 95 pallets per hour from the P&D locations[62].

Phase 2 (VNA Storing and picking of pallets from the APR)

- **Stage (i):** Automate the movement of pallets from the P&D location to the storage putaway in the VNA APR storage location[63]. This involves an automatic, unmanned VNA reach truck-type device that can scan labels and pick up pallets from the P&D area[64]. The aisle width is 2000 mm rack-to-rack, with a minimum of 1900 mm pallet-to-pallet[65].
- **Stage (ii):** Automate the reverse movement of pallets from the VNA APR storage location to the P&D location[66].

1.11 IT and power requirements and considerations

The supplier's proposal should indicate all required enabling works (power supplies, IT supplies, hardware, and software) and whether they will be supplied or are the responsibility of Firefly[67]. This includes:

- Positions and supply requirements for main control cabinets[68].
- Positions and electrical requirements for automation charging and other equipment[69].
- IT server requirements, data cabling routes, and access points[70].
- Integration and interfaces required with the Firefly Reflex WMS system[71].

1.12 Support and Maintenance

Any supplied automation solutions must include full support and maintenance packages[72]. All providers need to include a minimum four-week hypercare and training plan after the go-live dates[73]. Maintenance and service support proposals should be relevant to the site's working hours[74]. A maintenance contract specifying performance SLAs, a spares package, and all software and server support contracts must be included[75].

1.13 Clarifications and extra instructions

Proposals, costs, and benefits for Phase 1, Stages (i), (ii), and (iii) should be presented[76].

Phase 2 remains unchanged from the URS[77].

Item	Area	Phase 1 Tasks	Offered Y/N	Cost provided Y/N
A	Inbound	Automate the unloading of the trailers with internal conveyors (40% of total incoming at present).	No	No

B	Inbound	Consideration of improvements to the remainder of incoming loads (60%) to be auto unloaded or inducted to a conveyor system if manually unloaded.	No	No
C	Inbound	Automatic label scan and pallet board and load quality check and divert to a reject station for rework and reintroduction to the conveyor after remediation.	No	No
D	Inbound	Automatic transport BY CONVEYOR to P&D locations for pickup by VNA. No AGV required.	No	No
E	Inbound	Automatic transport BY AGV to P&D locations for pickup by VNA.	Yes	Yes
F	Outbound	Automate from P&D location after VNA deposit. AGV pickup and transport to manual floor marshall location.	Yes	Yes
G	Outbound	Automate from P&D location after VNA deposit. AGV pickup and transport/loading to conveyorized marshall lanes for loading to trailers.	No	No

H	Outbound	Automate from P&D location after VNA deposit. Auto conveyor feed to conveyorized marshall lanes for loading to trailers. No AGV required.	No	No
J	Outbound	Automate pickup by AGV from the P&D location and transport to marshalling location.	Yes	Yes
K	Outbound	Automate the loading of the outgoing trailer.	No	No